

STAT 24410/30030, Homework 7

Due Friday, December 6, at 10:00pm

1. (20 pts) Lecture Notes, Chapter 6, Exercise 1.

2. (20 pts) Lecture Notes, Chapter 6, Exercise 3.

3. (20 pts) Lecture Notes, Chapter 6, Exercise 4.

4. (20 pts) Lecture Notes, Chapter 6, Exercise 6.

5. (20 pts) In class, we talked about two ways of building credible intervals (you can also read about them on page 73 of Lecture Notes): equal-tailed and high-posterior density intervals. We assume here that the posterior distribution of θ is continuous with cumulative distribution function F and density function f . Assume (for simplicity) that f is unimodal with mode at m , and strictly monotone on $(-\infty, m)$ and on (m, ∞) . We build the following $(1 - \alpha)$ credible intervals:

(a) Equal-tailed: $[F^{-1}(\alpha/2), F^{-1}(1 - \alpha/2)]$.

(b) High posterior density: $I = \{\theta : f(\theta) \geq c\}$, where c is such that $P(\theta \in I) = 1 - \alpha$.

Show that the length of the interval in (b) is smaller or equal than the length of the interval in (a).